ATHANASIOS GOURDOMICHALIS
Java Programms
App4.java

The standard quadratic equation:

$$ax^2 + bx + c = 0$$

has two solutions given by the formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The first solution is obtained using the + sign instead of ±, and the second using the − sign.

Write an application that accepts values for a, b, and c (of **`type double`**) and then calculates the two solutions. If the quantity under the radical (discriminant) is a negative number, the equation has no real roots and your program should display an appropriate message. You may assume that the value of a is non-zero. Also, use any function from the Java **Math** class you need, for example to calculate the square root.

Example 1:

If the test input data is:

1.1 10.5 2.8

Then the expected output (solutions rounded to **3 decimal digits**) should be:

Enter the first number: **1,1**
Enter the second number: **10,5**
Enter the third number: **2,8**
The first solution is : -0,275
The second solution is: -9,271

Example 2:

If the test input data is:

1.0 1.0 10.0

Then the output should be:

Enter the first number: **1,0**
Enter the second number: **1,0**
Enter the third number: **10,0**
There are no real values for the quadratic equation.

Complete the Java code in the file **App4.java**

Every program must compile and run from the command line (not through an IDE).